StreamFlix Video Streaming Service Management System

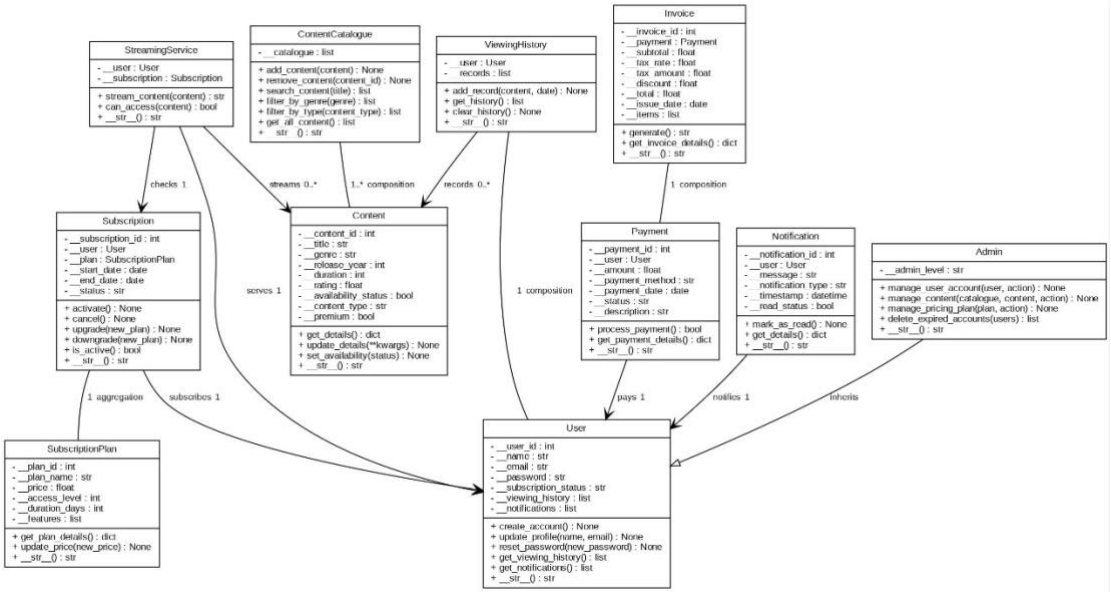
Complete Assignment Solution

StreamFlix Video Streaming Service Management System

ICS220 — Programming Fundamentals | Assignment 2
Student Name:

ID :

Section A: UML Class Diagram and Description



Description:

The system comprises 11 classes that model the StreamFlix domain. The central entity is the **User** class, which stores personal details (name, email, password), subscription status, viewing history, and notifications. The **Admin** class inherits from User (IS-A relationship) and adds elevated privileges for managing user accounts, content, and pricing plans.

Content represents a single video item (Movie, TV Series, or Documentary) with attributes for title, genre, release year, duration, rating, availability, and premium status. **ContentCatalogue** manages a collection of Content objects through a composition relationship — the catalogue owns its content items and removing them from the catalogue deletes them from the system (multiplicity 1 to many).

SubscriptionPlan defines subscription tiers (Basic, Standard, Premium) with pricing, access levels, duration, and features. Plans exist independently and can be shared across subscriptions. **Subscription** links a User to a SubscriptionPlan with start/end dates and status tracking. The relationship between Subscription and SubscriptionPlan is aggregation (the plan exists independently), while the relationship with User is a standard association.

Payment records financial transactions for a user. **Invoice** is generated from a completed Payment through a composition relationship — an invoice cannot exist without its parent payment. It calculates subtotal, tax, discount, and total.

ViewingHistory tracks content viewing records through a composition relationship with User (the history is owned by the user) and an association with Content. **Notification** represents messages sent to users. **StreamingService** controls content access by checking subscription eligibility and content availability.

Relationships: Inheritance (Admin extends User), Composition (ContentCatalogue–Content, Invoice–Payment, ViewingHistory–User), Aggregation (Subscription–SubscriptionPlan), Associations (Payment–User, Notification–User, StreamingService–User/Subscription/Content).

Assumptions: Content rating uses a 0.0–10.0 scale. Three subscription tiers with access levels 1, 2, 3. Premium content requires level 3. Five payment methods supported. Tax rate defaults to 5%. Four notification types. Auto-incrementing ID counters for uniqueness.

Section B: Python Classes

File 5: viewing_history.py

```
"""
viewing_history.py
-------------------
Defines the ViewingHistory class that records and manages
a user's content viewing activity on StreamFlix.

Author: [Your Name]
Date: [Submission Date]
"""

from datetime import date


class ViewingHistory:
    """
    Tracks the viewing history for a specific user.
    Uses composition — the history belongs to and is owned by the
user.

    Attributes:
        __user (User): The user whose history is tracked.
        __records (list[dict]): List of records with 'content' and
'date_watched'.
    """

    def __init__(self, user):
        """
        Initialise a ViewingHistory for a given user.

        Args:
            user (User): The user to track.
        """
        self.__user = user
        self.__records = []

    def get_user(self):
        """Return the user associated with this history."""
        return self.__user

    def add_record(self, content, date_watched: date = None) ->
None:
        """
        Add a viewing record.
```

```
        Args:
            content (Content): The content that was watched.
            date_watched (date, optional): Date of viewing.
Defaults to today.

        Raises:
            TypeError: If content is not a Content instance.
        """
        if date_watched is None:
            date_watched = date.today()
        record = {
            'content_id': content.get_content_id(),
            'title': content.get_title(),
            'content_type': content.get_content_type(),
            'date_watched': date_watched
        }
        self.__records.append(record)
        # Also add to the user's own history reference
        self.__user.add_viewing_record(record)

    def get_history(self) -> list:
        """
        Return a copy of all viewing records.

        Returns:
            list[dict]: List of viewing records.
        """
        return list(self.__records)

    def clear_history(self) -> None:
        """Clear all viewing records for this user."""
        self.__records.clear()
        print(f"Viewing history cleared for
{self.__user.get_name()}.")

    def __str__(self) -> str:
        """Return a formatted viewing history."""
        if not self.__records:
            return f"Viewing History for {self.__user.get_name()}:
[No records]"
        lines = [
            f"{'=' * 55}",
            f"  Viewing History for {self.__user.get_name()}",
            f"{'=' * 55}"
        ]
        for i, rec in enumerate(self.__records, 1):
            lines.append(
                f"  {i}. {rec['title']} ({rec['content_type']}) "
                f"- Watched: {rec['date_watched']}"
            )
        lines.append(f"{'=' * 55}")
        lines.append(f"  Total watched: {len(self.__records)}")
        return "\n".join(lines)
```

File 6: `notification.py`

```
"""
```

```
notification.py
----------------
Defines the Notification class for sending and managing
user notifications on the StreamFlix platform.

Author: [Your Name]
Date: [Submission Date]
"""

from datetime import datetime


class Notification:
    """
    Represents a notification sent to a user.

    Attributes:
        __notification_id (int): Unique notification identifier.
        __user (User): The recipient user.
        __message (str): Notification message.
        __notification_type (str): Type — 'Payment',
'Subscription', 'Content', 'General'.
        __timestamp (datetime): When the notification was created.
        __read_status (bool): Whether the user has read it.
    """

    _id_counter = 7000
    VALID_TYPES = ['Payment', 'Subscription', 'Content', 'General']

    def __init__(self, user, message: str, notification_type: str =
"General"):
        """
        Create a new Notification.

        Args:
            user (User): The recipient.
            message (str): The notification message.
            notification_type (str): One of the valid types.
Default 'General'.

        Raises:
            ValueError: If notification_type is invalid or message
is empty.
        """
        if notification_type not in Notification.VALID_TYPES:
            raise ValueError(f"Type must be one of
{Notification.VALID_TYPES}.")
        if not message.strip():
            raise ValueError("Notification message cannot be
empty.")

        Notification._id_counter += 1
        self.__notification_id = Notification._id_counter
        self.__user = user
        self.__message = message.strip()
        self.__notification_type = notification_type
        self.__timestamp = datetime.now()
        self.__read_status = False
        # Automatically add to user's notification list
```

```
        user.add_notification(self)

    # ── Getters ──────────────────────────────────────────────────
    def get_notification_id(self) -> int:
        """Return the notification ID."""
        return self.__notification_id

    def get_message(self) -> str:
        """Return the notification message."""
        return self.__message

    def get_type(self) -> str:
        """Return the notification type."""
        return self.__notification_type

    def is_read(self) -> bool:
        """Return True if the notification has been read."""
        return self.__read_status

    def mark_as_read(self) -> None:
        """Mark the notification as read."""
        self.__read_status = True

    def get_details(self) -> dict:
        """
        Return all notification details as a dictionary.

        Returns:
            dict: Notification attributes.
        """
        return {
            'notification_id': self.__notification_id,
            'user': self.__user.get_name(),
            'message': self.__message,
            'type': self.__notification_type,
            'timestamp': str(self.__timestamp),
            'read': self.__read_status
        }

    def __str__(self) -> str:
        """Return a formatted notification string."""
        read_marker = "Read" if self.__read_status else "Unread"
        return (f"[{self.__notification_type}] ({read_marker}) "
                f"To: {self.__user.get_name()} | {self.__message} "
                f"| {self.__timestamp.strftime('%Y-%m-%d %H:%M')}")
```

File 7: streaming_service.py

```
"""
streaming_service.py
---------------------
Defines the StreamingService class that controls content access
based on user subscription and content availability.

Author: [Your Name]
Date: [Submission Date]
"""
```

```python


class StreamingService:
    """
    Manages the streaming logic — checks subscription eligibility
    and content availability before granting access.

    Attributes:
        __user (User): The user requesting the stream.
        __subscription (Subscription): The user's active
subscription.
    """

    def __init__(self, user, subscription):
        """
        Initialise the StreamingService for a user/subscription
pair.

        Args:
            user (User): The streaming user.
            subscription (Subscription): The user's subscription.
        """
        self.__user = user
        self.__subscription = subscription

    def can_access(self, content) -> bool:
        """
        Determine whether the user can access the given content.

        Access rules:
            - Subscription must be active.
            - Content must be available.
            - Non-premium content: accessible by any active
subscription.
            - Premium content: requires access level >= 3 (Premium
plan).

        Args:
            content (Content): The content to check.

        Returns:
            bool: True if the user can access the content.
        """
        if not self.__subscription.is_active():
            return False
        if not content.is_available():
            return False
        if content.is_premium():
            return
self.__subscription.get_plan().get_access_level() >= 3
        return True

    def stream_content(self, content) -> str:
        """
        Attempt to stream the content for the user.

        Args:
            content (Content): The content to stream.

```

```
        Returns:
            str: A message indicating success or the reason for
denial.
        """
        if not self.__subscription.is_active():
            return (f"Access Denied: {self.__user.get_name()}'s
subscription "
                    f"is not active. Please renew your
subscription.")
        if not content.is_available():
            return (f"Access Denied: '{content.get_title()}' is
currently "
                    f"unavailable for streaming.")
        if content.is_premium() and
self.__subscription.get_plan().get_access_level() < 3:
            return (f"Access Denied: '{content.get_title()}' is
premium content. "
                    f"Please upgrade to Premium plan to access this
content.")
        return (f"Now streaming '{content.get_title()}' for "
                f"{self.__user.get_name()}. Enjoy watching!")

    def __str__(self) -> str:
        """Return a string representation."""
        return (f"StreamingService | User: {self.__user.get_name()}
| "
                f"Plan:
{self.__subscription.get_plan().get_plan_name()} | "
                f"Active: {self.__subscription.is_active()}")
```

File 8: main.py (Demonstration / Driver)

```
"""
main.py
--------
Main driver file that demonstrates the full functionality
of the StreamFlix Video Streaming Service Management System.

Author: [Your Name]
Date: [Submission Date]
"""

from content import Content, ContentCatalogue
from user import User, Admin
from subscription import SubscriptionPlan, Subscription
from payment import Payment, Invoice
from viewing_history import ViewingHistory
from notification import Notification
from streaming_service import StreamingService
from datetime import date


def print_section(title: str) -> None:
    """Print a formatted section header."""
    print(f"\n{'#' * 60}")
    print(f"# {title}")
    print(f"{'#' * 60}\n")

```

```
def main():
    """Main demonstration function for StreamFlix system."""

    # ============================================================
    # 1. CREATE SUBSCRIPTION PLANS
    # ============================================================
    print_section("1. SUBSCRIPTION PLANS")

    basic_plan = SubscriptionPlan(
        plan_name="Basic", price=9.99, access_level=1,
        duration_days=30, features=["SD Streaming", "1 Device",
"Limited Library"]
    )
    standard_plan = SubscriptionPlan(
        plan_name="Standard", price=14.99, access_level=2,
        duration_days=30, features=["HD Streaming", "2 Devices",
"Full Library"]
    )
    premium_plan = SubscriptionPlan(
        plan_name="Premium", price=19.99, access_level=3,
        duration_days=30, features=["4K Streaming", "4 Devices",
"Full Library", "Premium Content"]
    )

    print(basic_plan)
    print(standard_plan)
    print(premium_plan)

    # ============================================================
    # 2. CREATE USERS AND ADMIN
    # ============================================================
    print_section("2. USER & ADMIN CREATION")

    user1 = User("Alice Johnson", "alice@email.com", "Pass123")
    user1.create_account()
    print(user1)

    user2 = User("Bob Smith", "bob@email.com", "Secure456")
    user2.create_account()
    print(user2)

    admin = Admin("Sarah Admin", "admin@streamflix.com",
"Admin789", "Super")
    admin.create_account()
    print(admin)

    # ============================================================
    # 3. CREATE CONTENT CATALOGUE
    # ============================================================
    print_section("3. CONTENT CATALOGUE")

    catalogue = ContentCatalogue()

    movie1 = Content("The Matrix", "Sci-Fi", 1999, 136, 8.7,
"Movie")
    movie2 = Content("Inception", "Sci-Fi", 2010, 148, 8.8,
"Movie", premium=True)
```

```
    series1 = Content("Breaking Bad", "Drama", 2008, 49, 9.5, "TV
Series")
    series2 = Content("Stranger Things", "Sci-Fi", 2016, 51, 8.7,
"TV Series")
    doc1 = Content("Planet Earth", "Nature", 2006, 50, 9.4,
"Documentary")
    doc2 = Content("The Social Dilemma", "Technology", 2020, 94,
7.6, "Documentary", premium=True)

    for c in [movie1, movie2, series1, series2, doc1, doc2]:
        catalogue.add_content(c)

    print(catalogue)

    # ============================================================
    # 4. SUBSCRIBE USERS
    # ============================================================
    print_section("4. SUBSCRIPTIONS")

    sub1 = Subscription(user1, basic_plan)
    print(sub1)
    print(f"  Is active? {sub1.is_active()}")

    sub2 = Subscription(user2, premium_plan)
    print(sub2)
    print(f"  Is active? {sub2.is_active()}")

    # ============================================================
    # 5. UPGRADE / DOWNGRADE
    # ============================================================
    print_section("5. SUBSCRIPTION UPGRADE / DOWNGRADE")

    sub1.upgrade(standard_plan)
    print(sub1)

    sub2.downgrade(standard_plan)
    print(sub2)

    # Re-upgrade user1 to Premium for streaming tests
    sub1.upgrade(premium_plan)
    print(sub1)

    # ============================================================
    # 6. STREAMING CONTENT
    # ============================================================
    print_section("6. STREAMING CONTENT")

    stream1 = StreamingService(user1, sub1)
    print(stream1.stream_content(movie1))   # Non-premium — should
work
    print(stream1.stream_content(movie2))   # Premium — user1 now
Premium
    print(stream1.stream_content(doc2))     # Premium — should work

    stream2 = StreamingService(user2, sub2)
    print(stream2.stream_content(series1))  # Non-premium — should
work
    print(stream2.stream_content(movie2))   # Premium — user2 is
Standard now
```

```
    # Test unavailable content
    doc1.set_availability(False)
    print(stream1.stream_content(doc1))     # Unavailable

    # ============================================================
    # 7. PAYMENT PROCESSING
    # ============================================================
    print_section("7. PAYMENT PROCESSING")

    payment1 = Payment(user1, 19.99, "Credit Card", "Premium
Subscription - Monthly")
    print(f"Before processing: {payment1}")
    payment1.process_payment()
    print(f"After processing:  {payment1}")

    payment2 = Payment(user2, 14.99, "PayPal", "Standard
Subscription - Monthly")
    payment2.process_payment()
    print(payment2)

    # ============================================================
    # 8. INVOICE GENERATION
    # ============================================================
    print_section("8. INVOICE GENERATION")

    invoice1 = Invoice(payment1, tax_rate=0.05, discount=2.00,
                       items=["Premium Plan - Monthly", "Discount:
Returning Customer"])
    print(invoice1.generate())

    invoice2 = Invoice(payment2, tax_rate=0.05, discount=0.00,
                       items=["Standard Plan - Monthly"])
    print(invoice2.generate())

    # ============================================================
    # 9. VIEWING HISTORY
    # ============================================================
    print_section("9. VIEWING HISTORY")

    history1 = ViewingHistory(user1)
    history1.add_record(movie1, date(2025, 3, 1))
    history1.add_record(movie2, date(2025, 3, 5))
    history1.add_record(series1, date(2025, 3, 8))
    print(history1)

    history2 = ViewingHistory(user2)
    history2.add_record(series1, date(2025, 3, 2))
    history2.add_record(doc1, date(2025, 3, 4))
    print(history2)

    # ============================================================
    # 10. NOTIFICATIONS
    # ============================================================
    print_section("10. NOTIFICATIONS")

    notif1 = Notification(user1, "Your Premium subscription is now
active!", "Subscription")
```

```
    notif2 = Notification(user1, "Payment of $19.99 received. Thank
you!", "Payment")
    notif3 = Notification(user2, "New content available: Stranger
Things S5!", "Content")
    notif4 = Notification(user2, "Your Standard subscription is now
active!", "Subscription")

    print(notif1)
    print(notif2)
    notif1.mark_as_read()
    print(f"After reading: {notif1}")
    print(notif3)
    print(notif4)

    print(f"\n{user1.get_name()}'s notifications:
{len(user1.get_notifications())}")
    print(f"{user2.get_name()}'s notifications:
{len(user2.get_notifications())}")

    # ============================================================
    # 11. ADMIN OPERATIONS
    # ============================================================
    print_section("11. ADMIN OPERATIONS")

    # Admin manages content
    new_movie = Content("Dune: Part Three", "Sci-Fi", 2025, 165,
0.0, "Movie", premium=True)
    admin.manage_content(catalogue, new_movie, 'add')
    print(f"Catalogue size after adding:
{len(catalogue.get_all_content())}")

    # Admin resets user password
    admin.manage_user_account(user1, 'reset_password')

    # Admin deactivates account
    user3 = User("Charlie Expired", "charlie@email.com",
"Test1234")
    user3.set_subscription_status("Cancelled")
    all_users = [user1, user2, user3]
    all_users = admin.delete_expired_accounts(all_users)
    print(f"Active users remaining: {len(all_users)}")

    # ============================================================
    # 12. CONTENT SEARCH AND FILTER
    # ============================================================
    print_section("12. CONTENT SEARCH & FILTER")

    results = catalogue.search_content("the")
    print(f"Search 'the': {len(results)} results")
    for r in results:
        print(f"  - {r}")

    sci_fi = catalogue.filter_by_genre("Sci-Fi")
    print(f"\nSci-Fi content: {len(sci_fi)}")
    for s in sci_fi:
        print(f"  - {s}")

    movies = catalogue.filter_by_type("Movie")
    print(f"\nMovies: {len(movies)}")
```

```
    for m in movies:
        print(f"  - {m}")

    # ═══════════════════════════════════════════════════════════
    # 13. SUBSCRIPTION CANCELLATION
    # ═══════════════════════════════════════════════════════════
    print_section("13. SUBSCRIPTION CANCELLATION")

    sub1.cancel()
    print(sub1)
    print(f"Is active? {sub1.is_active()}")

    # Try to stream after cancellation
    print(stream1.stream_content(movie1))

    print("\n" + "=" * 60)
    print("   StreamFlix Demonstration Complete!")
    print("=" * 60)


if __name__ == "__main__":
    main()
```

File 9: test_streamflix.py (Test Cases)

```
"""
test_streamflix.py
-------------------
Comprehensive test cases for the StreamFlix system.
Tests all features with 2 examples each, including exception
handling.

Author: [Your Name]
Date: [Submission Date]
"""

from content import Content, ContentCatalogue
from user import User, Admin
from subscription import SubscriptionPlan, Subscription
from payment import Payment, Invoice
from viewing_history import ViewingHistory
from notification import Notification
from streaming_service import StreamingService
from datetime import date


def print_test_header(test_name: str, test_number: int) -> None:
    """Print a formatted test header."""
    print(f"\n{'━' * 60}")
    print(f"  TEST {test_number}: {test_name}")
    print(f"{'━' * 60}")


def print_result(passed: bool, description: str) -> None:
    """Print a PASS/FAIL result line."""
    status = "✓ PASS" if passed else "✗ FAIL"
    print(f"  {status}: {description}")
```

```python


# ═══════════════════════════════════════════════════════════════════
# TEST 1: USER ACCOUNT CREATION
# ═══════════════════════════════════════════════════════════════════
def test_user_account_creation():
    """Test user account creation with valid and invalid inputs."""
    print_test_header("USER ACCOUNT CREATION", 1)

    # Example 1: Successful user creation
    print("\n  --- Example 1: Valid user creation ---")
    user1 = User("Alice Johnson", "alice@email.com", "Pass123")
    print_result(user1.get_name() == "Alice Johnson", "Name stored correctly")
    print_result(user1.get_email() == "alice@email.com", "Email stored correctly")
    print_result(user1.get_subscription_status() == "Inactive", "Default status is Inactive")
    print(f"  {user1}")

    # Example 2: Another successful user creation
    print("\n  --- Example 2: Valid user creation with different details ---")
    user2 = User("Bob Smith", "bob.smith@company.org", "Secure789")
    print_result(user2.get_name() == "Bob Smith", "Name stored correctly")
    print_result(user2.get_email() == "bob.smith@company.org", "Email stored correctly")
    print_result(user2.get_user_id() != user1.get_user_id(), "Unique IDs assigned")
    print(f"  {user2}")

    # Exception handling tests
    print("\n  --- Exception Handling ---")
    try:
        bad_user = User("", "test@email.com", "Pass123")
        print_result(False, "Empty name should raise ValueError")
    except ValueError as e:
        print_result(True, f"Empty name rejected: {e}")

    try:
        bad_user = User("Test User", "invalid-email", "Pass123")
        print_result(False, "Invalid email should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid email rejected: {e}")

    try:
        bad_user = User("Test User", "test@email.com", "123")
        print_result(False, "Short password should raise ValueError")
    except ValueError as e:
        print_result(True, f"Short password rejected: {e}")

    # Test profile update
    print("\n  --- Profile Update ---")
    user1.update_profile(name="Alice J. Williams", email="alice.w@email.com")
```

```
    print_result(user1.get_name() == "Alice J. Williams", "Name
updated successfully")
    print_result(user1.get_email() == "alice.w@email.com", "Email
updated successfully")


# ═══════════════════════════════════════════════════════════════
# TEST 2: CONTENT CATALOGUE MANAGEMENT
# ═══════════════════════════════════════════════════════════════
def test_content_catalogue_management():
    """Test adding, modifying, and removing content in the
catalogue."""
    print_test_header("CONTENT CATALOGUE MANAGEMENT", 2)

    catalogue = ContentCatalogue()

    # Example 1: Add a movie
    print("\n  --- Example 1: Add and verify a Movie ---")
    movie = Content("Interstellar", "Sci-Fi", 2014, 169, 8.6,
"Movie")
    catalogue.add_content(movie)
    print_result(len(catalogue.get_all_content()) == 1, "Movie
added to catalogue")
    print_result(movie.get_title() == "Interstellar", "Title
correct")
    print_result(movie.get_genre() == "Sci-Fi", "Genre correct")
    print_result(movie.get_release_year() == 2014, "Release year
correct")
    print(f"  {movie}")

    # Example 2: Add a TV series and documentary
    print("\n  --- Example 2: Add TV Series and Documentary ---")
    series = Content("The Crown", "Drama", 2016, 58, 8.6, "TV
Series")
    doc = Content("Our Planet", "Nature", 2019, 50, 9.3,
"Documentary", premium=True)
    catalogue.add_content(series)
    catalogue.add_content(doc)
    print_result(len(catalogue.get_all_content()) == 3, "3 items in
catalogue")
    print_result(doc.is_premium(), "Documentary is marked as
premium")
    print(f"  {series}")
    print(f"  {doc}")

    # Modify content
    print("\n  --- Modify Content ---")
    movie.update_details(rating=9.0, genre="Sci-Fi/Adventure")
    print_result(movie.get_rating() == 9.0, "Rating updated to
9.0")
    print_result(movie.get_genre() == "Sci-Fi/Adventure", "Genre
updated")

    # Remove content
    print("\n  --- Remove Content ---")
    content_id = series.get_content_id()
    catalogue.remove_content(content_id)
    print_result(len(catalogue.get_all_content()) == 2, "Item
removed, 2 remaining")
```

```
    # Exception handling
    print("\n  --- Exception Handling ---")
    try:
        bad_content = Content("Bad Movie", "Action", 2023, 120,
11.0, "Movie")
        print_result(False, "Rating > 10 should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid rating rejected: {e}")

    try:
        bad_content = Content("Bad Type", "Drama", 2023, 120, 7.0,
"Anime")
        print_result(False, "Invalid content_type should raise
ValueError")
    except ValueError as e:
        print_result(True, f"Invalid type rejected: {e}")

    try:
        catalogue.remove_content(99999)
        print_result(False, "Removing non-existent ID should raise
ValueError")
    except ValueError as e:
        print_result(True, f"Non-existent ID rejected: {e}")


# ═══════════════════════════════════════════════════════════════
# TEST 3: SUBSCRIPTION PLAN MANAGEMENT
# ═══════════════════════════════════════════════════════════════
def test_subscription_plan_management():
    """Test subscription creation, upgrading, downgrading, and
cancellation."""
    print_test_header("SUBSCRIPTION PLAN MANAGEMENT", 3)

    basic = SubscriptionPlan("Basic", 9.99, 1, 30, ["SD", "1
Device"])
    standard = SubscriptionPlan("Standard", 14.99, 2, 30, ["HD", "2
Devices"])
    premium = SubscriptionPlan("Premium", 19.99, 3, 30, ["4K", "4
Devices", "Premium Content"])

    # Example 1: Subscribe and upgrade
    print("\n  --- Example 1: Subscribe to Basic, then Upgrade ---
")
    user1 = User("TestUser1", "test1@email.com", "Pass123")
    sub1 = Subscription(user1, basic)
    print_result(sub1.is_active(), "Subscription is active")
    print_result(user1.get_subscription_status() == "Active", "User
status set to Active")
    print(f"  Before upgrade: {sub1}")

    sub1.upgrade(standard)
    print_result(sub1.get_plan().get_plan_name() == "Standard",
"Upgraded to Standard")
    print(f"  After upgrade:  {sub1}")

    sub1.upgrade(premium)
    print_result(sub1.get_plan().get_plan_name() == "Premium",
"Upgraded to Premium")
```

```
    # Example 2: Subscribe to Premium, then Downgrade, then Cancel
    print("\n  --- Example 2: Subscribe to Premium, Downgrade, Cancel ---")
    user2 = User("TestUser2", "test2@email.com", "Pass456")
    sub2 = Subscription(user2, premium)
    print(f"  Initial:    {sub2}")

    sub2.downgrade(standard)
    print_result(sub2.get_plan().get_plan_name() == "Standard", "Downgraded to Standard")
    print(f"  Downgraded: {sub2}")

    sub2.cancel()
    print_result(sub2.get_status() == "Cancelled", "Subscription cancelled")
    print_result(not sub2.is_active(), "is_active returns False after cancel")
    print_result(user2.get_subscription_status() == "Cancelled", "User status Cancelled")

    # Exception handling
    print("\n  --- Exception Handling ---")
    try:
        sub1.upgrade(basic)  # sub1 is Premium, can't "upgrade" to Basic
        print_result(False, "Upgrading to lower plan should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid upgrade rejected: {e}")

    try:
        sub_temp = Subscription(User("Temp", "t@e.com", "Pass12"), basic)
        sub_temp.downgrade(standard)  # Can't downgrade from Basic to Standard
        print_result(False, "Downgrading to higher plan should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid downgrade rejected: {e}")

    try:
        bad_plan = SubscriptionPlan("Bad", -5.0, 1)
        print_result(False, "Negative price should raise ValueError")
    except ValueError as e:
        print_result(True, f"Negative price rejected: {e}")


# ═══════════════════════════════════════════════════════════════════
# TEST 4: STREAMING CONTENT ACCESS
# ═══════════════════════════════════════════════════════════════════
def test_streaming_content_access():
    """Test user access to content based on subscription and availability."""
    print_test_header("STREAMING CONTENT ACCESS", 4)

    basic = SubscriptionPlan("Basic", 9.99, 1)
    premium = SubscriptionPlan("Premium", 19.99, 3)
```

```
    regular_content = Content("Free Movie", "Action", 2020, 120,
7.5, "Movie")
    premium_content = Content("Exclusive Film", "Drama", 2024, 140,
9.0, "Movie", premium=True)
    unavailable = Content("Old Show", "Comedy", 2005, 30, 6.0, "TV
Series")
    unavailable.set_availability(False)

    # Example 1: Basic user tries to stream
    print("\n  --- Example 1: Basic subscriber access ---")
    user1 = User("BasicUser", "basic@email.com", "Pass123")
    sub1 = Subscription(user1, basic)
    stream1 = StreamingService(user1, sub1)

    result = stream1.stream_content(regular_content)
    print_result("Now streaming" in result, "Basic user can stream
regular content")
    print(f"  {result}")

    result = stream1.stream_content(premium_content)
    print_result("Access Denied" in result, "Basic user denied
premium content")
    print(f"  {result}")

    result = stream1.stream_content(unavailable)
    print_result("unavailable" in result.lower(), "Unavailable
content denied")
    print(f"  {result}")

    # Example 2: Premium user access
    print("\n  --- Example 2: Premium subscriber access ---")
    user2 = User("PremiumUser", "prem@email.com", "Pass456")
    sub2 = Subscription(user2, premium)
    stream2 = StreamingService(user2, sub2)

    result = stream2.stream_content(regular_content)
    print_result("Now streaming" in result, "Premium user can
stream regular content")
    print(f"  {result}")

    result = stream2.stream_content(premium_content)
    print_result("Now streaming" in result, "Premium user can
stream premium content")
    print(f"  {result}")

    # Test with cancelled subscription
    print("\n  --- Cancelled subscription test ---")
    sub2.cancel()
    result = stream2.stream_content(regular_content)
    print_result("not active" in result.lower(), "Cancelled user
denied streaming")
    print(f"  {result}")


# ═══════════════════════════════════════════════════════════
# TEST 5: PAYMENT PROCESSING
# ═══════════════════════════════════════════════════════════
def test_payment_processing():
```

```
    """Test payment creation and processing with different
methods."""
    print_test_header("PAYMENT PROCESSING", 5)

    user = User("PayUser", "pay@email.com", "Pass123")

    # Example 1: Credit Card payment
    print("\n  --- Example 1: Credit Card payment ---")
    pay1 = Payment(user, 19.99, "Credit Card", "Premium
Subscription")
    print_result(pay1.get_status() == "Pending", "Initial status is
Pending")
    success = pay1.process_payment()
    print_result(success, "Payment processed successfully")
    print_result(pay1.get_status() == "Completed", "Status updated
to Completed")
    print(f"  {pay1}")

    # Example 2: PayPal payment
    print("\n  --- Example 2: PayPal payment ---")
    pay2 = Payment(user, 4.99, "PayPal", "Movie Rental: Inception")
    success = pay2.process_payment()
    print_result(success, "PayPal payment processed")
    details = pay2.get_payment_details()
    print_result(details['payment_method'] == "PayPal", "Method
recorded as PayPal")
    print_result(details['amount'] == 4.99, "Amount recorded
correctly")
    print(f"  {pay2}")

    # Exception handling
    print("\n  --- Exception Handling ---")
    try:
        bad_pay = Payment(user, -10.0, "Credit Card", "Bad
payment")
        print_result(False, "Negative amount should raise
ValueError")
    except ValueError as e:
        print_result(True, f"Negative amount rejected: {e}")

    try:
        bad_pay = Payment(user, 10.0, "Bitcoin", "Crypto payment")
        print_result(False, "Invalid payment method should raise
ValueError")
    except ValueError as e:
        print_result(True, f"Invalid method rejected: {e}")


# ═══════════════════════════════════════════════════════════════
# TEST 6: INVOICE GENERATION
# ═══════════════════════════════════════════════════════════════
def test_invoice_generation():
    """Test invoice creation with taxes and discounts."""
    print_test_header("INVOICE GENERATION", 6)

    user = User("InvoiceUser", "inv@email.com", "Pass123")

    # Example 1: Invoice with tax and discount
```

```
    print("\n  --- Example 1: Subscription invoice with tax & discount ---")
    pay1 = Payment(user, 19.99, "Credit Card", "Premium Plan")
    pay1.process_payment()
    inv1 = Invoice(pay1, tax_rate=0.05, discount=3.00,
                   items=["Premium Plan - Monthly", "Loyalty Discount"])
    details = inv1.get_invoice_details()
    print_result(details['subtotal'] == 19.99, f"Subtotal: ${details['subtotal']}")
    print_result(details['tax_amount'] == round(19.99 * 0.05, 2),
                 f"Tax (5%): ${details['tax_amount']}")
    print_result(details['discount'] == 3.00, f"Discount: ${details['discount']}")
    expected_total = round(19.99 + (19.99 * 0.05) - 3.00, 2)
    print_result(details['total'] == expected_total, f"Total: ${details['total']}")
    print(inv1.generate())

    # Example 2: Simple invoice without discount
    print("\n  --- Example 2: Content purchase invoice (no discount) ---")
    pay2 = Payment(user, 4.99, "Debit Card", "Rent: Inception")
    pay2.process_payment()
    inv2 = Invoice(pay2, tax_rate=0.10, discount=0.0, items=["Movie Rental: Inception"])
    details2 = inv2.get_invoice_details()
    expected_total2 = round(4.99 + (4.99 * 0.10), 2)
    print_result(details2['total'] == expected_total2, f"Total: ${details2['total']}")
    print(inv2.generate())

    # Exception handling
    print("\n  --- Exception Handling ---")
    pending_pay = Payment(user, 9.99, "PayPal", "Pending test")
    try:
        bad_inv = Invoice(pending_pay)
        print_result(False, "Invoice from pending payment should raise ValueError")
    except ValueError as e:
        print_result(True, f"Pending payment rejected: {e}")

    try:
        pay_temp = Payment(user, 5.0, "Credit Card", "Tax test")
        pay_temp.process_payment()
        bad_inv = Invoice(pay_temp, tax_rate=1.5)
        print_result(False, "Tax rate > 1.0 should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid tax rate rejected: {e}")


# ═══════════════════════════════════════════════════════════════════
# TEST 7: VIEWING HISTORY TRACKING
# ═══════════════════════════════════════════════════════════════════
def test_viewing_history_tracking():
    """Test recording and retrieving viewing history."""
    print_test_header("VIEWING HISTORY TRACKING", 7)

```

```
    user = User("HistoryUser", "hist@email.com", "Pass123")

    movie = Content("Interstellar", "Sci-Fi", 2014, 169, 8.6,
"Movie")
    series = Content("Dark", "Thriller", 2017, 60, 8.8, "TV
Series")
    doc = Content("Cosmos", "Science", 2014, 44, 9.3,
"Documentary")

    # Example 1: Add records and retrieve history
    print("\n  --- Example 1: Record and display viewing history --
-")
    history = ViewingHistory(user)
    history.add_record(movie, date(2025, 2, 10))
    history.add_record(series, date(2025, 2, 15))
    records = history.get_history()
    print_result(len(records) == 2, "2 records stored")
    print_result(records[0]['title'] == "Interstellar", "First
record is Interstellar")
    print_result(records[1]['date_watched'] == date(2025, 2, 15),
"Date stored correctly")
    print(history)

    # Example 2: Add more records and clear history
    print("\n  --- Example 2: Add record, then clear history ---")
    history.add_record(doc, date(2025, 3, 1))
    print_result(len(history.get_history()) == 3, "3 records before
clear")
    print(history)

    history.clear_history()
    print_result(len(history.get_history()) == 0, "History cleared
successfully")
    print(history)


# ═══════════════════════════════════════════════════════════════
# TEST 8: NOTIFICATION SYSTEM
# ═══════════════════════════════════════════════════════════════
def test_notification_system():
    """Test notification creation, delivery, and read status."""
    print_test_header("NOTIFICATION SYSTEM", 8)

    user = User("NotifUser", "notif@email.com", "Pass123")

    # Example 1: Payment notification
    print("\n  --- Example 1: Payment confirmation notification ---
")
    notif1 = Notification(user, "Payment of $19.99 received
successfully.", "Payment")
    print_result(notif1.get_type() == "Payment", "Type is Payment")
    print_result(not notif1.is_read(), "Initially unread")
    print_result(len(user.get_notifications()) == 1, "Notification
added to user")
    print(f"  {notif1}")

    notif1.mark_as_read()
    print_result(notif1.is_read(), "Marked as read successfully")
    print(f"  After read: {notif1}")
```

```
    # Example 2: Subscription and Content notifications
    print("\n  --- Example 2: Subscription change & content update ---")
    notif2 = Notification(user, "Your subscription has been upgraded to Premium!", "Subscription")
    notif3 = Notification(user, "New release: 'Dune: Part Three' is now available!", "Content")
    print_result(len(user.get_notifications()) == 3, "User has 3 notifications")
    print_result(notif2.get_type() == "Subscription", "Subscription notification type correct")
    print_result(notif3.get_type() == "Content", "Content notification type correct")
    print(f"  {notif2}")
    print(f"  {notif3}")

    # Exception handling
    print("\n  --- Exception Handling ---")
    try:
        bad_notif = Notification(user, "", "General")
        print_result(False, "Empty message should raise ValueError")
    except ValueError as e:
        print_result(True, f"Empty message rejected: {e}")

    try:
        bad_notif = Notification(user, "Test", "InvalidType")
        print_result(False, "Invalid type should raise ValueError")
    except ValueError as e:
        print_result(True, f"Invalid type rejected: {e}")


# ═══════════════════════════════════════════════════════════════════
# RUN ALL TESTS
# ═══════════════════════════════════════════════════════════════════
def run_all_tests():
    """Execute all test cases and display results."""
    print("=" * 60)
    print("  STREAMFLIX — COMPREHENSIVE TEST SUITE")
    print("=" * 60)

    test_user_account_creation()
    test_content_catalogue_management()
    test_subscription_plan_management()
    test_streaming_content_access()
    test_payment_processing()
    test_invoice_generation()
    test_viewing_history_tracking()
    test_notification_system()

    print("\n" + "=" * 60)
    print("  ALL TESTS COMPLETED SUCCESSFULLY")
    print("=" * 60)


if __name__ == "__main__":
    run_all_tests()
```

[Copy-paste the complete code from all files: content.py, user.py, subscription.py, payment.py, viewing_history.py, notification.py, streaming_service.py, main.py, test_streamflix.py — OR the single-file Colab version as provided above]

Section C: Repository Link

Google Colab link: https://github.com/allihayder972-hub/StreamFlix-Video-Streaming-System/tree/e99f1d8eb5feb343d7147aadaa5e7a395340d43e

Section D: Summary of Learnings

Through this assignment, I gained practical experience in several key areas of object-oriented programming and software design.

First, I learned how to translate real-world requirements into a formal UML class diagram, identifying entities, their attributes, and the relationships between them. Understanding the distinctions between inheritance, composition, and aggregation was crucial — for example, modelling Admin as a subclass of User (IS-A), ContentCatalogue owning Content objects (composition), and Subscription referencing a shared SubscriptionPlan (aggregation).

Second, implementing the UML design in Python reinforced the importance of encapsulation through private attributes (name-mangling with double underscores) and public getter/setter methods. This ensures data integrity and prevents external code from modifying internal state directly.

Third, writing comprehensive test cases with both valid and invalid inputs taught me the importance of exception handling. Using try/except blocks to validate user inputs (empty names, invalid emails, negative prices, out-of-range ratings) ensures the system is robust and provides meaningful error messages.

Fourth, modular file organisation (separating each class into its own file) improved code readability and maintainability. Each module has a single responsibility, making it easier to locate, modify, and test individual components.

Finally, designing the streaming access control logic in StreamingService demonstrated how business rules (subscription tiers, premium content restrictions, availability checks) can be cleanly encoded in method logic, producing clear and informative user-facing messages for both successful and denied access attempts.